

Article

User Experience of Navigating Work Zones with Automated Vehicles: Insights from YouTube on Challenges and Strengths

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Highlights

What are the main findings?

- Automated vehicles (AVs) show a near-equal mix of strengths and challenges in navigating real-world construction zones, with 117 strengths and 105 challenges documented from 38 user-recorded cases.
- Tesla's FSD system demonstrates adaptability and dynamic response in complex work zones, while Waymo shows better handling of human interactions and structured environments.

What is the implication of the main finding?

- Continued AV development requires improvements in perception of temporary infrastructure, interaction with human actors, and navigation in unmapped or irregular layouts.
- User-generated video content offers valuable insights into real-world AV behavior, informing infrastructure design, policy development, and public trust in AV deployment.

Abstract

Understanding automated vehicle (AV) behavior in complex road environments and user attitudes in such contexts is critical for their safe and effective integration into smart cities. Despite growing deployment, limited public data exist on AV performance in construction zones; highly dynamic settings marked by irregular lane markings, shifting detours, and unpredictable human presence. This study investigates AV behavior in these conditions through qualitative, video-based analysis of user-documented experiences on YouTube, focusing on Tesla's supervised Full Self-Driving (FSD) and Waymo systems. Spoken narration, captions, and subtitles were examined to evaluate AV perception, decision-making, control, and interaction with humans. Findings reveal that while AVs excel in structured tasks such as obstacle detection, lane tracking, and cautious speed control, they face challenges in interpreting temporary infrastructure, responding to unpredictable human actions, and navigating low-visibility environments. These limitations not only impact performance but also influence user trust and acceptance. The study underscores the need for continued technological refinement, improved infrastructure design, and user-informed deployment strategies. By addressing current shortcomings, this research offers critical insights into AV readiness for real-world conditions and contributes to safer, more adaptive urban mobility systems.

Keywords: work zone; automated vehicles; transportation infrastructure; lane closure; urban planning; object detection; construction zone



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1. Introduction

Highway work zones in the United States have become increasingly hazardous for both workers and motorists. With numerous work zones established nationwide, transportation agencies are focusing on enhancing communication within these areas while also encouraging safer driving habits to reduce the risk of accidents [1]. Between 2013 and 2022, U.S. work zone fatalities rose by 50% [2]. In 2022 alone, crashes in these areas led to 891 deaths and 37,701 injuries. These incidents occurred within work zones or on roads leading to or from them due to traffic-related activities, behaviors, or controls. Of the 891 fatalities, 528 took place in construction zones, 305 were in work zones of unspecified type, 49 occurred in maintenance zones, and nine happened in utility zones [3]. The rapid advancement of automated vehicle (AV) technology holds great potential for transforming various aspects of urban and rural road transportation, including mobility, safety, and environmental impact under various road conditions [4,5]. According to the Society of Automotive Engineers (SAE), driving automation systems are classified into six levels, from Level 0 to Level 5, based on the degree of automation and the role of the human driver. Level 0 represents no automation, where the human driver performs all driving tasks. At Level 5, the system is fully automated, capable of handling driving tasks under all conditions without any human input. Levels 2 and 3 provide partial and conditional automation, respectively, where the vehicle can manage certain driving functions, but the human driver must remain attentive and ready to intervene. Levels 4 and 5 eliminate the need for driver intervention in specific (Level 4) or all (Level 5) driving environments. This taxonomy is essential for understanding the functional expectations and limitations of AV systems operating in complex scenarios such as work zones [6]. Work zones are expected to remain a challenging environment for automated vehicle (AV) systems due to their dynamic and unpredictable nature. These zones often involve temporary changes to lane configurations, vary in duration (ranging from short-term to long-term), may affect a single lane or multiple lanes, and can be present year-round. Additionally, irregular signage, pavement markings, fluctuating speed limits, and the presence of construction equipment and workers require AVs to exhibit heightened awareness and adaptability to navigate safely [1].

Numerous research studies have evaluated and are still assessing the performance, benefits, and challenges of connected and/or automated vehicles in urban and rural transportation settings using field tests, real-world data, simulations, surveys and interviews [7,8], or based on user experience on video sharing platforms [9]. Various datasets exist for AVs, including disengagement and incident data from the California Department of Motor Vehicles [10,11], NHTSA AV incident data [12], and Waymo's diverse automated driving datasets [13]. However, there is a significant lack of studies employing any systematic method aimed at assessing AV performance and rider experience, specifically in work zone environments; consequently, a shortage of publicly available datasets on this topic is observed. This is a crucial gap, as AVs are expected to account for a notable share of vehicles in U.S. cities by 2030 [14]. Understanding the challenges and potential of AVs in work zones, along with public perception, acceptance, and willingness to ride in AVs or along with AVs despite the possibility of encountering construction zones, is essential for guiding infrastructure owners and operators on necessary physical and digital modifications. This will inform manufacturers on required AV technology development and assist policymakers in establishing regulations that support the safe and efficient integration of AVs in future urban and rural environments, ensuring that individuals across different financial backgrounds, geographic locations, physical abilities, technological literacy levels,

and health conditions not only have the economic means but also feel confident and willing to consider AVs as a viable transportation option.

This paper analyzes the performance of AVs navigating work zones, drawing on real-world case studies shared by users on YouTube between 2018 and 2024. The analysis focuses on strengths and challenges related to safety, mobility, and reliability to identify areas where AV technology and infrastructure can improve to better support future deployment. The study examines how two major AV systems, Tesla's Full Self-Driving (FSD) and Waymo, responded to diverse construction environments. It is worth noting that throughout this study, any reference to FSD refers to "supervised" Full Self-Driving systems. According to the SAE definitions of autonomy levels, there are currently no commercially available Level 5 vehicles; none that can operate under all conditions without human oversight or restriction.

Observations are organized into six key categories: Perception & Sensing, Decision Making & Planning, Control & Speed Management, Human Interaction & Override, Infrastructure Compliance & Navigation, and Environmental & Weather Conditions. In addition to technical performance, the study offers insights into user experience and perception. It explores how AV disengagements and errors are interpreted by different road users, including passengers, pedestrians, construction workers, and human drivers, and how these responses reflect moments of trust, frustration, and stress. The findings help evaluate whether AV systems like supervised FSD reduce the cognitive burden on drivers and riders in complex work zones or inadvertently intensify it. They also reveal whether users are more likely to trust the system or initiate disengagement, seek remote assistance, or even prefer human-driven vehicles in these situations. This study is guided by the following research questions:

- (1) What types of challenges do AV users report encountering when navigating construction zones, and what strengths do they demonstrate in such complex environments?
- (2) To what extent do current AV technologies meet user expectations regarding safety, reliability, and automated performance in work zones, and do the strengths outweigh the challenges?
- (3) How do users respond to AV behavior in construction zones, and what do their reactions reveal about trust, acceptance, and willingness to adopt the technology in these contexts?
- (4) What insights can be extracted from user-shared videos to guide future improvements in AV technology, transportation systems, urban infrastructure, and policy development related to AV deployment in work zones?

2. Materials and Methods

This study adopts a qualitative exploratory approach, incorporating elements of a scoping review to analyze user-shared YouTube videos and examine how AVs navigate construction zones, while also assessing user opinions about AV performance. The analysis focuses on videos recorded by individuals who were either AV users (riders/passengers) or external observers, such as drivers or passengers in nearby vehicles, or pedestrians who captured AV behavior in real-world settings. These publicly accessible videos provide valuable insights into AV performance, human-AV interaction, and user perception that are often absent from formal datasets or require the integration of multiple research methodologies to fully uncover.

The analysis process began by developing a series of keyword combinations aligned with the research objectives to find videos containing terms such as "Work Zone" or "Construction Zone" in the title, in conjunction with at least one keyword representing automated driving behavior, including "self-driving," "automated vehicle," "automated

car,” or “autopilot.” Table 1 outlines the conceptual categories and associated synonyms and terms used in constructing the search queries. For selecting AV manufacturers, we focused on companies that are recognized as major players in the AV industry and have conducted frequent public road testing. Since the goal of this study was to compare AV performance and user experiences over time, we targeted companies for which user-documented experiences were available consistently across multiple years and covered various software versions, updates, and geographic locations across different U.S. states and, in some cases, other countries. Based on these criteria, Waymo and Tesla were selected as the primary focus in this study. Other AV manufacturers were considered during the video screening process. However, these companies did not have a sufficient volume of user-generated content that met our inclusion criteria.

Table 1. Key concepts and synonyms used for search query development.

Concepts	Synonyms/Related Terms
Automated vehicle	Self-driving car, automated vehicle, automated car, driverless vehicle
Work zone	Construction zone, roadwork, road construction
Navigation challenges	Lane shift, obstacle, detour, cones, temporary signage
Major AV companies with public road testing	Tesla, Waymo

In the initial phase of our methodology, video acquisition was executed through a purpose-written Python 3.11.9 script that interacts with the official YouTube Data API v3. For each search cycle, the script iterated through a predefined set of Boolean keyword strings (listed in Table 1) and submitted each query to the API with the “publishedAfter” and “publishedBefore” parameters set to 1 January 2018 and 31 December 2024, respectively. The upload date range was selected to capture both early and recent AV deployments, providing a longitudinal view of technological evolution and public response.

The API returned the title, publication date, and video identifier for each query. Two case-insensitive regular expressions were then applied: one confirming the presence of work-zone terminology (construction zone, work zone, roadwork), and the other ensuring reference to an AV concept (automated vehicle, self-driving, Tesla, Waymo, automated vehicle, or full self-driving). Items satisfying both patterns were retained, and a URL-based check was used to remove duplicates arising from overlapping queries. No restrictions on video duration were applied, allowing inclusion of all content types (including standard videos and shorts). As a result, the selected videos ranged from less than one minute to over 20 min in length. Next, all extracted videos were individually watched in full to ensure their relevance to the topic and alignment with the study’s criteria and objectives. Some videos resulting from the search were manually excluded from the analysis, despite being titled as demonstrations of AV navigation in work zones and have relevant contents. The excluded videos met one or more of the following criteria: (1) they were released by AV companies, which were omitted to focus exclusively on user-generated content and ensure an unbiased representation of real-world experiences; (2) they lacked narration, captions, or subtitles, and were excluded to avoid potential misinterpretation by the authors. To facilitate the summarization and accurate interpretation of AV behavior and user experiences, the selected videos were then categorized into two groups for analysis: (1) those containing spoken narration (supported by captions, descriptions, or subtitles), and (2) those without spoken audio, relying on written captions or descriptions. For videos and shorts that included only on-screen text or descriptions without spoken audio, all

content was read, analyzed, and summarized. For videos with spoken narration, transcripts were extracted using YouTube’s built-in “Show transcript” feature to access the official, time-stamped dialog. Each line of speech was reviewed for emotional cues, such as confidence, hesitation, surprise or frustration, and descriptive comments regarding AV performance. User sentiment was inferred through a multimodal qualitative approach that combined three layers of analysis: verbal content, paralinguistic cues, and behavioral observations. Verbal content included emotionally charged phrases such as “I don’t feel safe” or “that was smooth.” Paralinguistic cues such as tone of voice, inflection, pacing, and audible signals like laughter or tension were used to interpret emotional states with greater accuracy. Behavioral observations, including hand movements, facial expressions, or manual disengagements, were also considered when available to support or clarify the emotional tone of the narration. A calm voice paired with a positive comment indicated confidence, while strained speech and physical signs of discomfort reflected anxiety or loss of trust. This approach ensured a comprehensive and reliable analysis across different content types.

In the next step, thematic analysis was conducted with the assistance of OpenAI’s API to identify key themes related to the strengths and challenges presented in the collected references. This approach aligns with recent research that utilizes advanced large language models (LLMs) to analyze AV disengagement reports, demonstrating the models’ effectiveness in identifying patterns, classifying causes, and informing infrastructure improvements based on complex, unstructured data [15]. Original transcripts and video summaries were programmatically analyzed to extract recurring patterns, with the API supporting pattern recognition and clustering, from which categories and subcategories of AV performance were developed.

To assess the accuracy of the model’s thematic categorization of AV strengths and challenges, we conducted a manual evaluation using the full set of extracted events. From the dataset of 222 events (117 strengths and 105 challenges) identified across 38 user-shared videos, all events were manually reviewed and categorized based on their root causes through detailed analysis of video content, captions, and subtitles. We then compared the manually assigned root cause category to the one assigned by the model. A prediction was considered correct if the model’s assigned category matched the manual classification. Out of the 117 strength events, the model’s categorization matched the manual classification in 104 cases. For the 105 challenge events, the model matched the manual classification in 93 cases. This results in an overall accuracy of:

$$\text{Accuracy} = (\text{Number of Correct Predictions} / \text{Total Evaluated Events}) \times 100$$

$$\text{Accuracy} = ((104 + 93) / (117 + 105)) \times 100 = 88.67\%$$

This indicates that the model correctly categorized approximately 89% of the evaluated events. This process led to the establishment of six main thematic categories: Perception & Sensing, Decision Making & Planning, Control & Speed Management, Human Interaction & Override, Infrastructure Compliance & Navigation, and Environmental & Weather Conditions.

3. Results

This section synthesizes 38 user-documented experiences involving two AV companies, Waymo and Tesla, between 2018 and 2024. Figure 1 illustrates the distribution of YouTube video content related to AV navigation in work zones during this period. From 2018 to 2021, the percentage of available English-language video content remained relatively low and stable, with each year contributing approximately 5% of the overall distribution. This suggests that, in the early stages, AV navigation in work zones was either less frequently documented or had not yet become a major public focus. A notable rise occurred in

2020 with the share increasing to about 16%, likely reflecting expanded AV testing and growing efforts to demonstrate performance under complex conditions. Although a slight decline followed in 2021, content volume rebounded in 2022 to around 13%, coinciding with the resumption of broader testing activities after pandemic-related slowdowns. The most significant growth was observed in 2023, which accounted for nearly 29% of the total content, highlighting intensified AV activity in construction zones and reflecting technological advancements, heightened regulatory interest, and greater public scrutiny. In 2024, while the share slightly decreased to approximately 26%, it remained substantially higher than in earlier years, indicating sustained engagement and interest. The growth in video content underscores the increasing importance of evaluating AV performance in unpredictable work zone environments as the technology continues to mature.

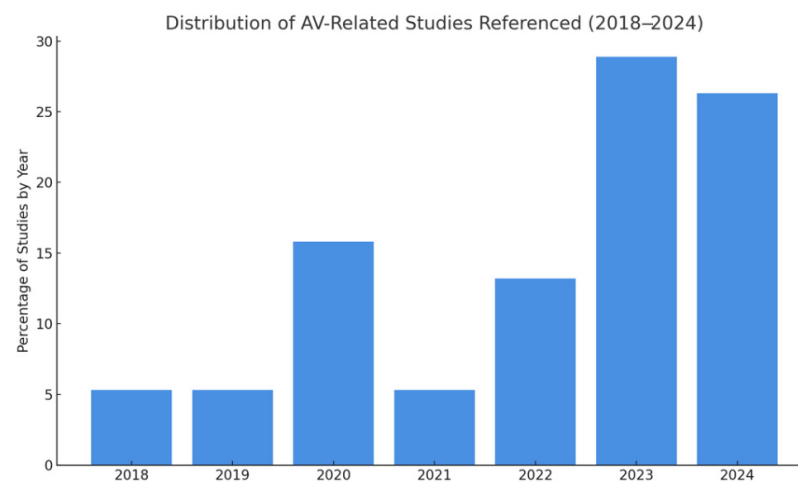


Figure 1. Trend of studies on AV deployment in construction zones (2018–2024).

The synthesis is organized into two subsections, beginning with Waymo and followed by Tesla. Within each subsection, user experiences are arranged chronologically based on the video upload dates, allowing for an examination of how user perceptions and AV system performance have evolved over time. Synthesizing user-generated content on YouTube played a critical role in revealing both the progress and persistent challenges of AV performance in complex and dynamic construction zone environments. These videos offer authentic glimpses into AV behavior under real-world conditions, capturing moments of user confidence, discomfort, hesitation, and fear that simulation studies, surveys, and interviews alone often fail to replicate. An important feature of this study is the broad appeal of its content. The insights captured in the results section are valuable to a wide audience, including transportation professionals and industry specialists seeking technical observations, as well as general users who are interested in gaining a better understanding of AV performance in real-world settings.

3.1. Waymo's Navigation Experience

Waymo is an automated driving technology company and a subsidiary of Alphabet Inc., Google's parent company. Established in 2009 as the Google Self-Driving Car Project, it was rebranded as Waymo in 2016. The company's mission is to make it safe and easy for people and goods to move around. Waymo develops the "Waymo Driver," the world's first autonomous ride-hailing service that combines hardware and software, including LiDAR, cameras, radar, and a powerful artificial intelligence (AI) computer platform, to provide a 360-degree view of the driving environment. This technology aims to enable Waymo's vehicles to navigate complex urban environments without human intervention. Waymo continues to expand its services and test its technology in various environments to improve

road safety and redefine mobility [16]. Waymo's performance in construction zones, as observed through user-recorded videos, reflects a blend of technological advancements and persistent challenges.

In 2021, a Waymo vehicle in Chandler, Arizona, encountered difficulty navigating around construction cones placed at an intersection. The vehicle came to a stop and requested remote assistance. However, similar difficulties occurred later, indicating limited capability in adapting to temporary construction barriers [17]. In contrast, Waymo demonstrated notable improvement in 2022. During a test in downtown Phoenix, Arizona, the AV successfully maneuvered through road barriers, navigated around jaywalkers, and operated effectively in wet conditions. The trip, which was arranged via the Waymo app, was completed without major incidents, showcasing the system's growing ability to handle complex urban conditions [18]. However, the year 2023 introduced new concerns. In Tempe, Arizona, one Waymo vehicle accelerated from a stop to 50 mph within a construction zone where the posted speed limit was 35 mph. It failed to stop at a red light and exited the work zone traveling at 45 mph, which raised questions about the system's compliance with temporary speed and signal regulations [19]. That same year in Los Angeles, California another Waymo vehicle entered two separate work zones. Although it used turn signals correctly, it became confused by abrupt lane changes and required remote assistance to navigate safely. Additionally, dense fog during the drive complicated the vehicle's ability to interpret its surroundings, though it ultimately avoided any unsafe behavior. Despite these challenges, some 2023 tests highlighted the system's strengths [20]. In one instance, a Waymo vehicle navigated around active construction equipment in the main road without human intervention, earning praise from passengers [21]. In another real-world encounter, Waymo vehicle was observed driving into an active construction site without any human occupants. As it approached the area, the vehicle seemed confused by the irregular layout, including an open trench and scattered construction cones. Its perception system failed to identify a clear path, and the vehicle came to a stop near the edge of the trench, unable to determine how to proceed. The situation drew the attention of bystanders, who noted the vehicle's hesitation and lack of response. To redirect it, a construction worker placed cones in front of the vehicle, which eventually prompted it to reverse and steer away from the hazard [22]. During a set of tests in San Francisco that same year, the Waymo vehicle encountered multiple construction and obstruction-related events in an urban setting. It began its route by preparing to make a right turn onto a street obstructed by cones and a stationary motorcycle. The cones and signage were clearly visible on the passenger display. The vehicle maintained steady behavior as traffic progressed slowly due to a temporary stop sign held by a construction crew. Once the sign changed to "slow," the vehicle responded appropriately and continued forward. Further along the route, the AV approached another zone impacted by utility or road maintenance work. Instead of continuing straight as initially planned, the system automatically rerouted with a smooth right turn, seamlessly re-entering the flow of traffic. Later, it encountered a double-parked vehicle with an open door and cones surrounding it. Despite the limited space, the vehicle identified the hazard and executed a precise maneuver around the obstruction. This moment even drew a smile from a nearby pedestrian who witnessed the AV's careful behavior. Toward the end of the drive, the vehicle adjusted its route again in response to ongoing tree maintenance. Arborists had placed cones and equipment along the lane, prompting the system to detect the obstruction and alter its path accordingly. Throughout this journey, the AV's actions remained steady, predictable, and minimally disruptive, contributing to the user's positive impression of its ability to manage real-world, cone-dense environments with composure [23]. Nonetheless, in 2024, Waymo experienced another setback. In Chandler, California, a Waymo vehicle became immobilized within a work zone. Nearby construction

workers attempted to assist by waving and moving cones to create a clearer path. Despite these efforts, the vehicle remained unresponsive and unable to proceed, highlighting an ongoing weakness in dealing with unexpected layout changes and a lack of clear external guidance [24]. Table 2 presents a summary of Waymo performance and user perceptions in various construction zone conditions between 2021 and 2024, highlighting instances of major disengagements or assistance and the corresponding overall user sentiment.

Table 2. Waymo performance and user perception in construction zones.

Year	Location	Work Zone Condition	Major Disengagement/ Assistance	Waymo Overall User Perception
2021 [17]	Chandler, Arizona	Construction cones at an intersection.	Yes	Negative
2022 [18]	Phoenix, Arizona	Barriers, jaywalkers, wet conditions, red light.	No	Positive
2023 [19]	Tempe, Arizona	Posted speed limit of 35 mph.	No	Negative
2023 [20]	Los Angeles, California	Required abrupt lane changes, light rain and dense fog.	Yes	Mixed
2023 [21]	N/A	Active construction equipment.	No	Positive
2023 [22]	N/A	Active construction site with irregular layout, open trench, scattered cones, presence of construction workers.	Yes	Negative
2023 [23]	San Francisco, California	Multiple work zones, street cones, stationary motorcycle, temporary stop sign held by a construction crew, signage change from “Stop” to “Slow,” double-parked vehicle with an open door, presence of pedestrian, ongoing tree maintenance site, construction equipment.	No	Positive
2024 [24]	Chandler, California	Unexpected layout changes, cones, presence of construction workers.	Yes	Negative

Figure 2a,b present word clouds visualizing user-reported challenges and strengths of Waymo in work zones, respectively, highlighting key themes in negative and positive feedback.



Figure 2. (a,b): Word Clouds of user-reported challenges and strengths of Waymo in work zones.

3.2. Tesla's Navigation Experience

Tesla, founded in 2003, is a global leader in electric vehicles and clean energy, aiming to accelerate the transition to sustainable energy [23]. In addition to producing EVs, Tesla develops Autopilot and supervised Full Self-Driving (FSD) technologies, which use a vision-based system powered by cameras, neural networks, and AI to perceive and interpret driving environments. Although these systems require active driver supervision,

they are continuously refined through over-the-air updates and large-scale data collection [25]. Since 2018, users have tested Tesla's systems in real-world construction zones, highlighting both progress and challenges. Early tests showed that Autopilot could follow cones when lane markings were absent [26] and use aggressive settings like "Mad Max" to manage merges effectively [27]. In 2019, Autopilot version 2019.40.1.1 handled obstructed lanes by smoothly changing lanes without visualizing the barrels on-screen [28], and another test demonstrated its ability to maintain stable speed and direction through cone-defined paths [29]. In 2020, Autopilot handled rainy construction zones well at first but struggled when lane visibility declined, triggering emergency braking and requiring driver intervention [30]. Another Model 3 test showed strong lane-following in tight zones, aided by visualization updates, though merging still required driver readiness [31]. Autopilot 2020.16.2.1 showed improved rendering of signs and cones but failed at speed adjustment and made unsafe lane attempts in construction-diverted areas [32]. A comprehensive 2020.4.1 test highlighted both near-collisions with cones and later success in fully navigating an unmarked, cone-only lane; even using a lead vehicle as a dynamic guide or performing the route solo [33]. Autopilot version 2020.40.3 represented a leap in capability with improved object detection, night vision, and traffic light recognition in city work zones [34]. It also handled complex maneuvers across solid lines and cones with high precision [35]. By 2021, FSD Beta 10.5 could stop workers holding signs but misinterpreted movement cues, prompting multiple driver interventions [36]. In 2022, version 10.69.2.2 struggled in blocked-lane scenarios and required manual control [37], while FSD 0.2.2 failed consistently in a simple cone-lined path despite succeeding in highway conditions [38]. Other users noted jerky corrections, wide turns, and hesitation under debris-strewn conditions, though the system remained stable [39]. FSD Beta 10.11.2, tested in 2022, performed well through a complex construction route with cyclists, stop signs, and ambiguous intersections, requiring brief driver input [40]. In 2023, testing in Dallas showed FSD identifying a lane closure, slowing to 30 mph, and executing a clean lane change [41]. Other tests that year revealed better cooperation with human drivers [42], stable barrier navigation [43], and dynamic human awareness, such as safe clearance during tree trimming operations [44]. However, the system still exhibited uncertainty, hugging lane edges near potholes [45], and occasionally drove too slowly, requiring manual throttle input to keep pace with traffic [46]. In 2024, the release of FSD version 12.3.3 marked significant advancements. In complex work zones with shifting cones and signage, the system struggled initially but adjusted well by following a lead vehicle and mimicking human-like behavior with only minor intervention [47]. Another test showed confident adaptation to construction cones and lane shifts [48], and Beta 12 rerouted automatically at a fenced closure, succeeding where Beta 11 had failed [49]. Version 12.3.6 also handled uneven pavement, sticker-only lane markings, tight merges, and heavy cone density with composure, correcting its own mistaken lane entries without disengagement [50]. A separate test in mixed rain conditions showed the system reducing speed from 77 to 70 mph and managing misaligned signs, construction cones, and parking without intervention [51]. A Cybertruck test saw the system reroute around a road closure and self-park while recognizing pedestrians and adapting to residential speeds [52]. Tesla's Basic Autopilot also showed reliable stop-and-go control and predictive behavior in adaptive traffic near temporary lane closures [53]. Near Arthur Ashe Stadium, FSD 12.3.6 successfully scanned traffic gaps and executed a smooth lane change through a narrow work zone, demonstrating enhanced real-time decision-making [54]. Despite these gains, a real-world 2024 test revealed recurring limitations in urban construction zones: misinterpreting inactive signals, failing to identify closures, poor rerouting, hesitation at intersections, and visual inconsistencies in cone detection; all contributing to the need for frequent driver intervention [55]. Table 3

presents a comprehensive summary of user-documented evaluations of Tesla Autopilot and supervised Full Self-Driving (FSD) systems from 2018 to 2024, detailing vehicle models and software versions tested, construction zone conditions, observed outcomes, major disengagements events, and users' overall perceptions.

Table 3. Tesla performance and user perception in construction zone.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2018 [26]	Model 3	Construction zone without lane markings, used cones.	Adapted to atypical conditions centered itself based on cones.	No	Positive
2018 [27]	Model X on Mad Max mode	Nighttime construction, Mad Max Lane changes, dynamic layout.	- Executed smooth lane shifts and merged into traffic. - Used turn signals automatically and enhanced driver awareness with UI cues. - Hesitated occasionally during lane changes and struggled in tight merges. - Prioritized caution over aggressiveness, even in assertive mode.	No	Positive
2019 [28]	Autopilot 2019.40.1.1 (HW2.5)	Construction barrels obstructing lanes, limited visibility.	- Performed smooth lane change to maintain a safe buffer. - Demonstrated cautious but effective behavior without object visualization. - Detected and reacted to barrels. - Transitioned back to the original lane after passing the obstruction.	No	Positive
2019 [29]	Model 3 (HW 3.0)	Construction zone with cones, no visible lane markings.	- Maintained consistent speed. - Followed temporary boundaries created by cones instead of painted pavement markings.	No	Positive
2020 [30]	Autopilot 2020.8.1	Rainy construction zone, low visibility, emergency braking triggered.	- Detected construction cones. - Adjusted speed appropriately. - Maintained lane position in light traffic. - Navigated the initial portion of the work zone smoothly. - Followed the flow of surrounding traffic. - Exhibited instability as the roadway narrowed. - Struggled with reduced visibility and poorly marked lanes. - Drifted toward the edge of the road near a reflective barrier. - Triggered emergency braking system to avoid impact.	Yes	Mixed
2020 [31]	Model 3	Unclear road markings, cones present.	- Demonstrated strong lane-following performance. - Maintained lane integrity using cameras and sensors. - Detected nearby obstacles. - Used enhanced visualization updates to detect cones and construction elements. - Struggled with lane changes in unpredictable sections. - Faced difficulty merging in complex work zone layouts. - Required ongoing driver readiness.	No	Positive

Table 3. Cont.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2020 [32]	Autopilot 2020.16.2.1	Motorway, roundabout in UK, construction zones with cones and temporary markings.	- Failed to change lanes automat- edly on a dual carriageway. - Attempted to enter a closed lane diverted by cones. - Hesitated around large vehicles sometimes misjudging safe gaps for lane changes. - Executed abrupt lateral shifts at junctions. - Failed to adjust speed to temporary speed limits in construction zones. - Aborted a turn unexpectedly in a car park.	Yes	Mixed
2020 [33]	Autopilot 2020.4.1 (HW 3.0)	Complex construction zone, cones, newly paved unmarked lanes.	- Executed a lane shift. - Veered dangerously close to a reflector cone, nearly struck the vehicle's mirror. - Triggered a strong driver reaction and increased caution. - Displayed a 40-mph speed despite a 35-mph construction zone. - Adjusted speed later in response to conditions, slowed down noticeably in cone-dense areas, even if traffic moved faster. - Followed another car through the unmarked zone, as a dynamic guide. - Navigated the freshly paved lane using only cone placement. - Made the correct turn and maintained proper spacing. - Demonstrated capability for fully visual-based navigation, surprised and impressed the user.	No	Positive
2020 [34]	Autopilot 2020.40.3	Urban nighttime construction zones, narrow shifting lanes.	- Recognized cyclists. - Navigated through narrow and shifting construction lanes. - Used high-resolution cameras for object identification. - Executed careful and cautious lane changes in dynamic environment. - Detected signal changes from 150 to 200 m away. - Came to a complete stop when needed. - Required driver confirmation to proceed through intersections. - Integrated real-time map data. - Made accurate navigational decisions even when road cues were missing or unclear.	No	Positive
2020 [35]	Model 3 2020.40.3	A work zone, where cones directed the vehicle to cross a solid center line in construction.	- Narrowly avoided a cone by less than one foot. - Stayed within the temporary cone-defined path. - Smoothly transitioned back to the original lane after passing the obstruction. - Demonstrated high precision. - Showed strong spatial awareness and precision. - Received user praise for handling the challenge effectively.	No	Positive

Table 3. Cont.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2021 [36]	Model 3 FSD Beta 10.5	Construction-heavy area in Houston, Texas, worker with stop sign, potholes	- Recognized and responded to a worker holding a stop sign, came to a safe stop. - Misinterpreted the worker's slight forward movement as a signal to proceed. - Attempted to follow the stop sign holder, requiring driver intervention. - Upon re-engagement, tried to steer around the worker, prompting another manual takeover. - Unexpectedly followed the walking worker with the stop sign. - Disengaged when a pedestrian entered the intersection. - Struggled to handle a T-intersection without occasional disengagements. - Maintained cautious behavior near barricades and construction workers. - Had difficulty on pothole-ridden roads, requiring frequent driver takeover. - Failed to slow down at a railroad crossing; driver had to apply brakes.	Yes	Mixed
2022 [37]	Model Y FSD Beta 10.69.2.2	Nighttime, construction cones, blocked lane, confusion at intersections.	- Navigated basic turns smoothly, handled quiet streets without issue. - Became confused at an intersection, made a wrong turn. - Hesitated and failed to identify a navigable path, when encountered cones and a blocked lane in a construction zone, came to a full stop.	Yes	Negative
2022 [38]	FSD	Five consecutive trials through a temporary right-hand lane defined by construction cones, slight uphill grade with clear lane marking and consistent cone placement	- Hesitated at the approach to cones in each trial. - Backed up slightly at the fourth cone during every run. - Stalled and failed to proceed past the cones. - Misinterpreted a construction worker's "Slow" sign gesture as a cue to steer toward a cone. - Repeated failures occurred despite clear and consistent conditions. - Performed reliably in structured freeway environments (executed 85 mph lane changes, operated well even in heavy traffic and nighttime conditions).	Yes	Negative

Table 3. Cont.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2022 [39]	FSD	Gravel-strewn lane, uneven surfaces, cautious navigation.	- Briefly hesitated before navigating the lane. - Made a small, jerky steering correction. - Proceeded cautiously through the gravel-strewn path. - Made wide turns at intersections. - Occasionally drifted near curbs. - Responded slowly to changes in speed. - A stable and safe passage due to conservative driving behavior. - Successfully navigated uneven terrain and scattered debris. - Resumed normal operation outside the work zone without further incident.	No	Positive
2022 [40]	FSD Beta 10.11.2	Active construction zone with stop signs, cyclists, intersections, and traffic congestion, low visibility drone mounted on the vehicle captured the drive	- Navigated complex routes confidently. - Smoothly handled a notoriously confusing intersection by following another vehicle. - Demonstrated adaptive perception and interpreted an ambiguous trailer correctly. - Made well-timed directional decisions, even under pressure from tailgating vehicles. - Maintained control in low-visibility conditions with unclear lane markings.	No	Positive
2023 [41]	FSD	Construction zone in Dallas, Texas with lane closure and residential navigation	- Identified a lane closure. - Automatically reduced speed to 30 mph. - Executed a smooth lane change to avoid the obstruction. - Performed a slightly abrupt right turn into a residential neighborhood.	No	Positive
2023 [42]	FSD	Narrowed road section due to construction	- Approached a narrowed road section safely, yielded to oncoming traffic. - Exchanged a visual acknowledgment with a human driver (indicative of cooperative behavior). - Maintained a smooth trajectory throughout the interaction. - Sustained reliable lane positioning. - Ensured consistent obstacle detection under physical and social driving conditions.	No	Positive
2023 [43]	FSD Beta	Construction zone with barriers.	- Navigated around construction barriers with stable control. - Adjusted its path safely and smoothly. - Maintained appropriate following distances.	No	Positive

Table 3. Cont.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2023 [44]	Model 3 FSD Beta	Tight construction zone in Silicon Valley, California with human workers and construction equipment trimming trees	- Displayed strong situational awareness. - Dynamically adjusted to the presence of humans and roadside equipment. - Maintained safe clearance from the active work area.	No	Positive
2023 [45]	Model 3	Road narrowing with large pothole	- Exhibited uncertainty as the road narrowed. - Created its own lane by closely hugging the right edge. - Approached a large pothole unsafely.	Yes	Negative
2023 [46]	Model Y FSD Beta	Cone-dense construction zone, conservative speed (California).	- Recognized and reacted to each cone. - Maintained overly cautious speed. - Required light manual acceleration by the driver to maintain appropriate traffic flow.	No	Positive
2024 [47]	Model 3 FSD 12.3.3	Complex work zone with shifting lanes, scattered cones, and temporary signage.	- Struggled to interpret faded or missing lane markings. - Had difficulty with unexpected obstacles early in the test. - Identified and began reliably following a lead vehicle. - Quickly adjusted its trajectory to match the traffic flow. - Exhibited human-like decision-making (adjusted speed and path). - Required only minimal driver intervention (Inputs were limited to occasional steering corrections or system acknowledgments).	No	Positive
2024 [48]	Model 3 FSD 12.3.3	Construction zone with cones and lane shifts	- Demonstrated impressive performance in construction zones, addressed challenges that traditional autopilot systems struggled with. - Adapted to dynamic scenarios. - Executed smooth lane changes despite construction obstacles like cones and lane shifts.	No	Positive
2024 [49]	FSD Beta 12	Fenced-off construction area with road closure and adjacent equipment	- Immediately recognized the road closure and nearby equipment. - Steered into and followed the detour without hesitation. - Adjusted correctly to altered lane markings. - Maintained a steady and controlled pace through the work zone.	No	Positive

Table 3. Cont.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2024 [50]	FSD 12.3.6	Dynamic construction zone, uneven pavement, cone density.	• Detected need for a lane change due to road work well in advance, executed smooth lane change despite moderate surrounding traffic. • Crossed a double yellow line confidently. • Navigated uneven pavement. • Maintained proper spacing around cones and curbs. • Stayed within path marked by temporary stickers (not paint), with only minor wobble. • Responded properly at a red light and proceeded smoothly on green. • Adjusted trajectory effectively around new curbs and tight traffic sections. • Managed areas with no lane lines on the navigation map using real-time perception. • Adapted to lane shifts and merged seamlessly with traffic. • Maintained safe clearance in narrow, cone-dense zones with roadside activity. • Mistakenly entered a left-turn lane but corrected smoothly and re-entered the appropriate lane. • Successfully navigated fake turn lanes, obstructive equipment, and tight lane shifts. • Reacted appropriately to audible alerts from nearby objects. • Maintained composure and continued operation without hesitation.	No	Positive
2024 [51]	FSD 12.3.6	City streets, highways, and an active construction zone; Light, intermittent rain	• Operated with minor performance warnings despite visibility issues from rain. • Adjusted speed from preset 77 mph to 70 mph in response to weather conditions. • Encountered a new construction layout with cones and yellow lane markings, recognized ongoing drainage work. • Hesitated at a stop line with triangular markings, required a tap on the accelerator to proceed through heavy cross-traffic. • Managed through misaligned stop signs and construction cones. • Merged efficiently into a tight traffic gap, adjusted speed according to surrounding vehicle flow. • Paused at a red right-turn arrow until prompted by driver tap (due to vehicle pressure behind).	No	Positive

Table 3. Cont.

Year	Vehicle Model/ Software Version	Work Zone Condition	Outcome	Major Disengage- ment/Assistance	Tesla User Overall Perception
2024 [52]	Cybertruck FSD	Road closure due to construction on the main road	- Detected the road closure on the main road, rerouted automatically through local streets. - Encountered workers and traffic cones on the alternate route, slowed down appropriately near active work. - Recognized nearby pedestrians and vehicles. - Adapted well to residential neighborhood conditions by maintaining appropriate speeds, adjusted braking naturally.	No	Positive
2024 [53]	Model 3 Autopilot	Urban driving in Pittsburgh, Pennsylvania with stop-and-go traffic near temporary lane closures in active construction areas.	- Monitored behavior of nearby vehicles continuously, and adjusted its own speed based on surrounding traffic dynamics. - Maintained a safe following distance. - Reduced overall driver workload. - Demonstrated predictive modeling for enhanced comfort and safety.	No	Positive
2024 [54]	FSD 12.3.6	Complex, constrained construction work zone in Queens, New York City.	- Scanned for an available gap in traffic or lane, executed a smooth lane change to the left within a construction zone. - Successfully navigated through challenging and constrained road conditions.	No	Positive
2024 [55]	FSD	Urban construction zone with complex (inactive signals, road closures, construction barriers)	- Misinterpreted inactive traffic signals as green lights. - Drove 8–10 mph below posted speed limits. - Failed to recognize road closures and construction barriers. - Struggled with real-time reroute, caused confusion and unsafe maneuvers. - Occasionally entered repeated loops in dead ends. - Hesitated at stop signs and blinking red lights. - Creeped forward without confidently determining right-of-way. - Misidentified objects inconsistently displayed cones and construction elements. - Handled basic lane keeping well in clearly marked areas. - Performance was unreliable in construction-heavy scenarios.	Yes	Negative

Figure 3a,b present word clouds visualizing user-reported challenges and strengths of Tesla in work zones, respectively, highlighting key themes in negative and positive feedback.

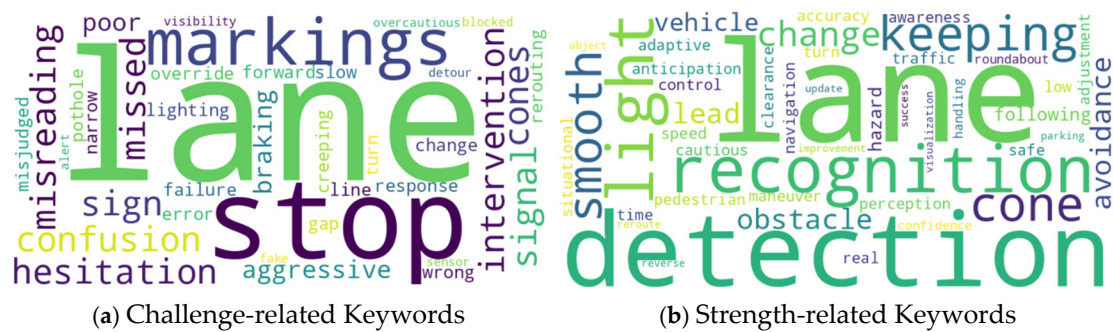


Figure 3. (a,b): Word Clouds of user-reported challenge and strengths of Tesla in work zones.

4. Discussion

4.1. AV Performance Analysis

Analyzing the synthesis of 38 video contents, 222 events were identified, comprising 105 challenges and 117 strengths. This near-even split, with strengths representing 52.7% and challenges 47.3% of the total, reflects a transitional phase in AV development, where systems are becoming increasingly capable but still require further refinements to achieve consistently reliable performance when navigating dynamic and unpredictable work zone environments. Thematic analysis of user-documented encounters with AVs in construction environments led to a deeper understanding of distinct challenge and strength categories along with their distribution, as shown in Figure 4.

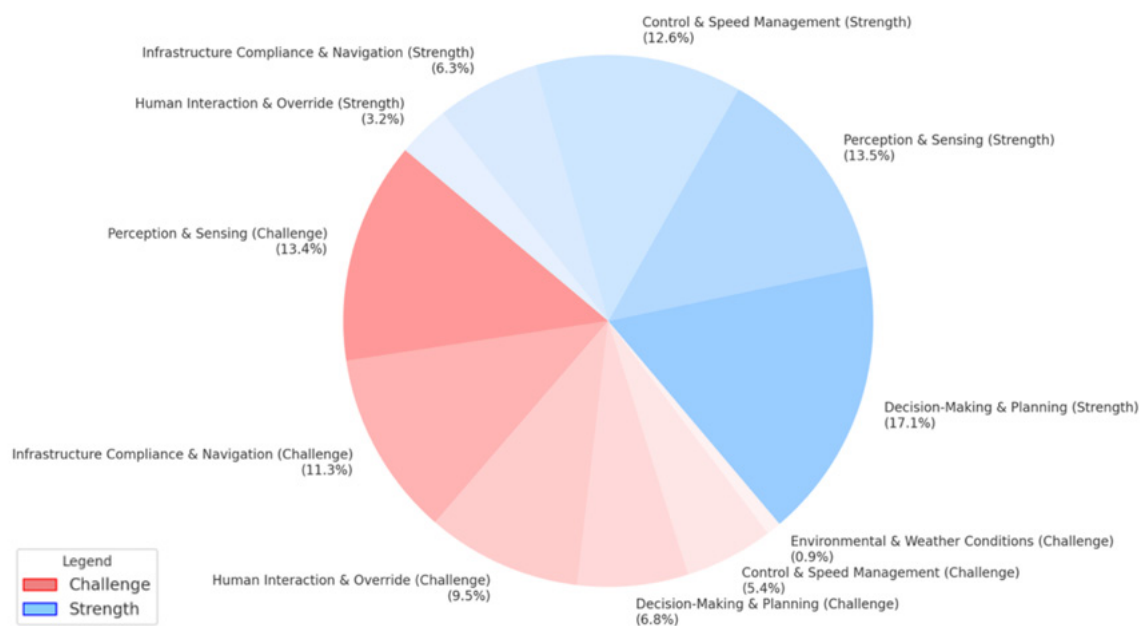


Figure 4. Distribution of strength and challenges in AV work zone navigation.

4.1.1. Challenges

In Table 4, the challenge categories and subcategories are ranked by frequency, with Perception & Sensing emerging as the most reported issue, followed by Infrastructure Compliance & Navigation, and Human Interaction & Override. These results indicate that perception-related challenges remain a primary concern as AVs often struggle to detect, interpret, and respond to temporary, unexpected, or uncommon road features such as construction signs, cones, and irregular lane markings. Infrastructure compliance also presents a significant challenge, with vehicles failing to recognize traffic signals, respond to missing or unclear lane guidance, or properly navigate detours. Furthermore, difficul-

ties in human interaction, such as responding appropriately to pedestrians, construction workers, and other unpredictable human behaviors, contribute to disengagement events. Although fewer challenges were reported in Decision-Making & Planning, Control & Speed Management, and Environmental & Weather Conditions, occasional failures in these areas suggest that further advancements are needed to ensure reliable AV operation in dynamic and complex construction environments. In particular, the low frequency of events related to environmental and weather conditions is likely attributed to the limited availability of real-world testing conducted under adverse weather scenarios.

Table 4. Ranking and total number of reported AV work zone navigation challenges.

Rank	Challenge Category	Challenge Subcategories	Tesla	Waymo	Total Frequency
1	Perception & Sensing	- Non-standard Markings; - Temporary Signs; - Occluded Visibility; - Unexpected Obstacles.	28	2	30
2	Infrastructure Compliance & Navigation	- Traffic Signals; - Construction Zone Entry and Exit; - Unclear/Missing Lane Markings.	19	6	25
3	Human Interaction & Override	- Pedestrian/Worker Interaction; - System Disengagements/Human Intervention Required.	17	4	21
4	Decision-Making & Planning	- Complex Intersection/Turning movement; - Lane Prediction/Planning; - Road Closures or Detours.	13	2	15
5	Control & Speed Management	Speed Modulation.	11	1	12
6	Adverse Weather Conditions	Presence of Precipitation/Fog.	1	1	2
All	All	All	89	16	105

A comparative breakdown of challenge categories for 38 user-recorded videos involving Tesla and Waymo (Figure 5) reveals key differences in the operational performance of the two companies.

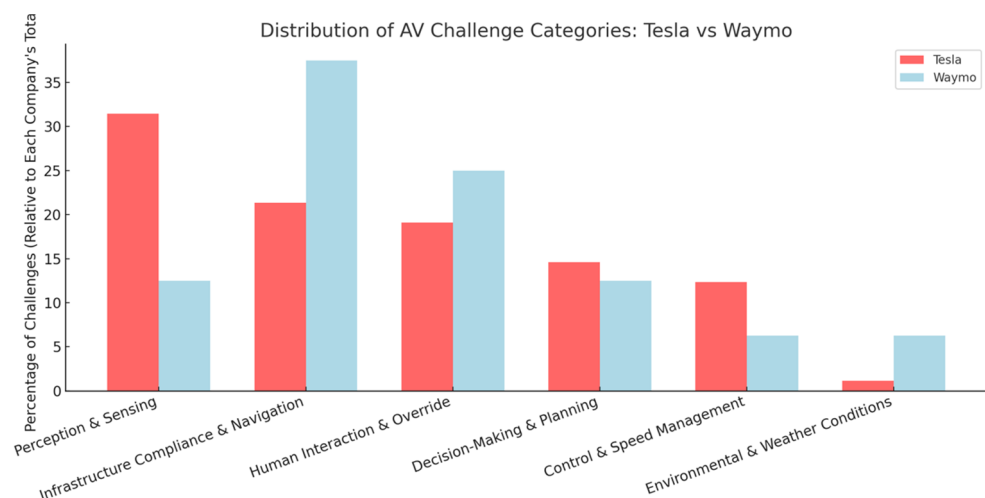


Figure 5. Comparison of AV challenge types encountered by Tesla and Waymo.

Tesla showed a higher rate of perception and sensing challenges (31.46%) compared to Waymo (12.50%), likely due to Tesla's camera-only system versus Waymo's multi-sensor fusion using LiDAR, radar, and cameras. Waymo's richer sensor suite appears more robust in detecting and interpreting the driving environment. Conversely, Waymo faced more issues with infrastructure compliance and navigation (37.50%) than Tesla (21.35%), especially in poorly marked roads. Tesla's neural-network-based system seems better at generalizing to dynamic or unmapped environments, while Waymo's map-dependent planning may struggle in these contexts. Human override needs were present in both systems (Waymo 25%, Tesla 19.10%). Decision-making and planning issues were slightly more common in Tesla (14.61%) than in Waymo (12.50%). Similarly, speed or control challenges occurred more frequently in Tesla (12.36%) than in Waymo (6.25%), suggesting that Tesla exhibited less smooth control under certain conditions. Weather-related events were rare overall, with only one reported incident for each company.

4.1.2. Strengths

In Table 5, the strength categories highlight key areas where AVs demonstrated notable performance in construction zones. Decision-Making & Planning emerged as the most notable strength, driven by the AVs' ability to follow lead vehicles near construction zones, reroute around obstacles, and adapt to dynamic environments. Following lead vehicles near construction zones particularly reflected human-like decision-making, allowing AVs to navigate through complex and changing conditions. Perception & Sensing ranked next, with AVs demonstrating improved capabilities in detecting obstacles and interpreting road signage, although occasional inconsistencies still highlight areas for further enhancement. Control & Speed Management also showed strong performance as AVs were able to execute smooth lane changes in detours and adjust speed cautiously near cones and other temporary elements. Lower rankings were observed in Infrastructure Compliance & Navigation, where strengths included lane re-centering and estimation, and in Human Interaction & Override, where AVs demonstrated some ability to handle pedestrian interactions. The relatively lower frequencies in these categories suggest that, while progress has been made, further development is necessary to achieve consistent and reliable AV behavior in construction environments.

Table 5. Ranking and total number of AV work zone navigation strengths.

Rank	Strength Category	Strength Subcategories	Tesla	Waymo	Total Frequency
1	Decision-Making & Planning	- Following Lead Vehicle; - Rerouting and Path Planning; - Adaptation to Dynamic Environment.	32	6	38
2	Perception & Sensing	- Road Signs; - Obstacles.	27	3	30
3	Control & Speed Management	- Smooth Lane Change in Detours; - Slowdown/Cautious Speed near Cones.	25	3	28
4	Infrastructure Compliance & Navigation	Lane Re-centering and Estimation.	13	1	14
5	Human Interaction & Override	- Handling of Pedestrian Interactions; - No Major Human Intervention.	5	2	7
Total	All	All	102	15	117

The comparative strength distribution of Tesla and Waymo across five categories (Figure 6) reveals important distinctions when interpreted alongside their corresponding challenge frequencies. These distributions highlight not only where each system excels, but also how those strengths relate to observed limitations in construction zone navigation.

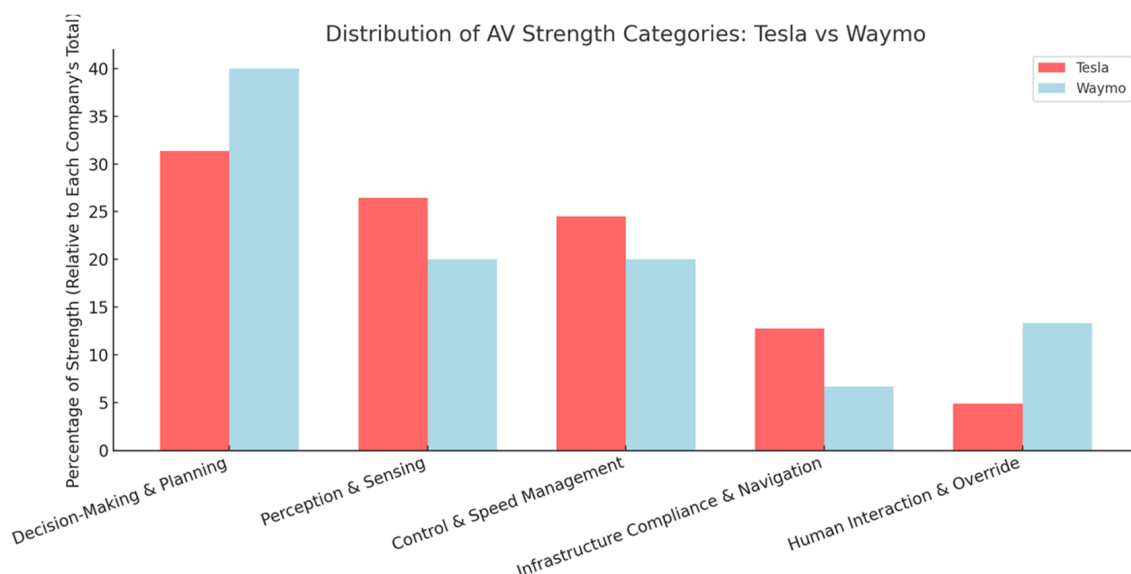


Figure 6. Distribution of strength categories for Tesla and Waymo in work zones.

Tesla demonstrates a broader distribution of strengths, with notable performance in Perception & Sensing (26.47%) and Control & Speed Management (24.51%). These align with Tesla's camera-based system, which frequently succeeded in identifying cones, signs, and other road elements while maintaining cautious speed adjustments. Interestingly, these areas were also among the most challenging categories for Tesla; Perception & Sensing (31.46%) and Control & Speed Management (12.36%); suggesting that while Tesla encounters issues in these domains, it has also shown repeated improvements and system adaptability in response. This dual presence of strengths and challenges reflects Tesla's iterative learning model, where real-world data informs software updates that gradually improve performance. Tesla's strength in Infrastructure Compliance & Navigation (12.75%) also aligns with a relatively moderate challenge rate (21.35%), indicating that the system, despite some navigation difficulties in construction layouts, often successfully handles temporary infrastructure changes using cone guidance and visual cues. Meanwhile, Decision-Making & Planning (31.37%) is another strong area for Tesla, supported by real-world examples of smooth detours and dynamic responses, although its challenge rate in this category (14.61%) suggests occasional hesitation or misjudgment in high-stakes environments. However, Human Interaction & Override represents Tesla's weakest area of strength (4.90%) and corresponds with a notably high challenge frequency (19.10%). This can be primarily attributed to Tesla's supervised Full Self-Driving (FSD) system that assumes that a human will intervene when necessary, unlike Waymo, which is built to operate without driver input. Waymo's strengths are more concentrated, with the majority falling under Decision-Making & Planning (40%) and Human Interaction & Override (13.33%). These strengths correspond well to Waymo's low challenge percentages in Decision-Making (12.5%) and Control (6.25%), showing that its rule-based, map-supported system excels in structured planning and cautious interaction with pedestrians and construction workers. However, Waymo's greatest challenge area, Infrastructure Compliance & Navigation (37.5%), is also where it shows the least strength (6.67%), pointing to a significant gap in adapting to temporary or unmapped construction layouts. Similarly, its relatively low

strength in Perception & Sensing (20%) reflects modest ability to respond fluidly to visual changes, which matches a moderate challenge rate (12.5%) in that category.

4.1.3. Comparison Between Strengths and Challenges

A comparative analysis of overall strength and challenge categories also reveals notable trends in the evolving performance of AV systems within construction zone environments (Figure 7). Decision-making & Planning demonstrated the greatest improvement, with 38 strength events recorded compared with 15 challenge events. While AVs occasionally struggled with complex intersections as well as lane and detour prediction, the substantially higher number of strengths suggests that capabilities in rerouting, path planning, and adaptation to dynamic environments have matured considerably. In contrast, Perception & Sensing presented a more balanced outcome, with both strengths and challenges recorded in 30 events each. Although AVs have demonstrated improved abilities in obstacle detection and road sign recognition within work zones, persistent difficulties in interpreting temporary signs, uncommon markings, occluded visibility, and unexpected obstacles highlight ongoing inconsistencies in perception. This may indicate two key insights: perception and sensing remain a central focus of AV performance, representing the most frequently observed category with a total of 60 events; and user feedback on AVs' perception and sensing capabilities in work zones is evenly divided. Control & Speed Management exhibited a favorable trend, with 28 strength events reported versus 12 challenge events, indicating that AVs are increasingly capable of managing speed modulation and executing smooth lane changes under variable conditions. However, Infrastructure Compliance & Navigation continues to present notable challenges, with 25 challenge events recorded compared with 14 strengths related to lane re-centering and estimation. Human Interaction & Override also remain critical areas for improvement, with 21 challenge events and only seven strength events reported. This highlights the complexity of responding effectively to pedestrian interactions and unexpected human behavior and emphasizes that human intervention, either by the driver or through remote assistance, is still necessary in certain situations. Environmental & Weather conditions were cited infrequently, with only two challenge events and no corresponding strengths, suggesting limited exposure to adverse conditions in the summarized YouTube content.

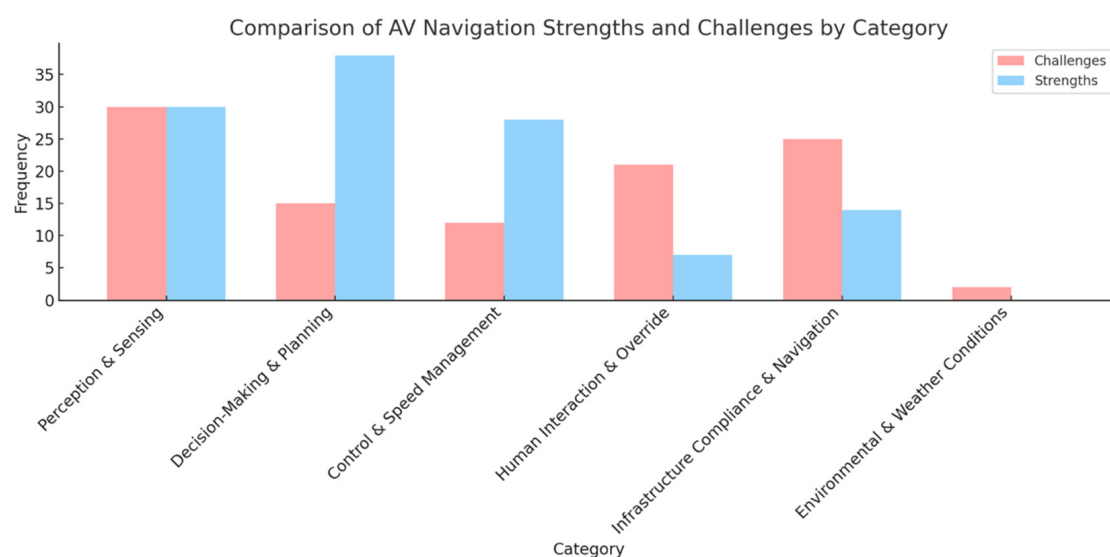


Figure 7. Comparison of challenges and strengths of AV navigation in work zones.

4.2. User Perception Analysis

The analysis of user-documented experiences reveals an overall positive perception of AVs navigating work zones, although notable concerns persist. Among the 38 experiences analyzed between 2018 and 2024, approximately 74% of user perceptions were positive, 11% were mixed, and 15% were negative (Figure 8).

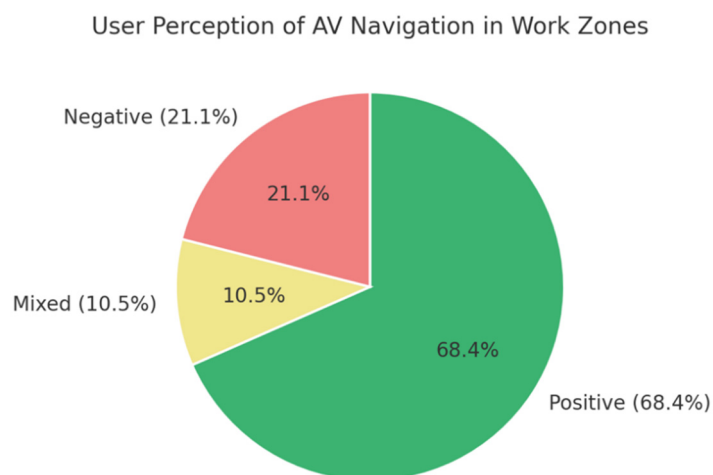


Figure 8. Overall user perception of AV navigation in work zones.

Users frequently praised AVs' strengths in structured tasks such as smooth lane changes, cautious speed modulation, and obstacle avoidance in dynamic construction environments. However, as Tables 2 and 3 showed, negative and mixed perceptions were associated with cases where major disengagements or external interventions were necessary. Specifically, negative or mixed feedback was often reported when AVs encountered issues such as confusion with temporary infrastructure layouts (e.g., open trenches and scattered cones), failure to correctly interpret construction signs or lane closures, hesitation or incorrect responses to pedestrians and construction workers, difficulties navigating poorly marked lanes under low-visibility conditions, and abrupt or unsafe maneuvers requiring manual takeover. These events highlight that while AV systems like Tesla's supervised FSD and Waymo's driverless technology have made progress in controlled scenarios, challenges in interpreting complex and rapidly changing construction environments continue to affect user trust and acceptance.

An analysis of the user perception chart comparing Tesla and Waymo (Figure 9) further illustrates these dynamics. Among Tesla's recorded experiences (30 events), approximately 77% were positive, 13% negative, and 10% mixed, indicating a predominantly favorable response from users. In contrast, Waymo's user feedback (8 events) was more divided, with 38% positive, 50% negative, and 13% mixed perceptions. Waymo's higher proportion of negative responses, despite the smaller sample size, may reflect either more challenging test conditions or more frequent system limitations during complex navigation tasks. This disparity may also be partly attributed to operational differences: Waymo's system functions in a fully automated mode without a human driver present at the wheel, whereas Tesla's supervised FSD trials involve a safety driver capable of intervening. The absence of a human driver in Waymo vehicles may heighten user sensitivity to system behavior, especially in uncertain or high-risk environments like work zones.

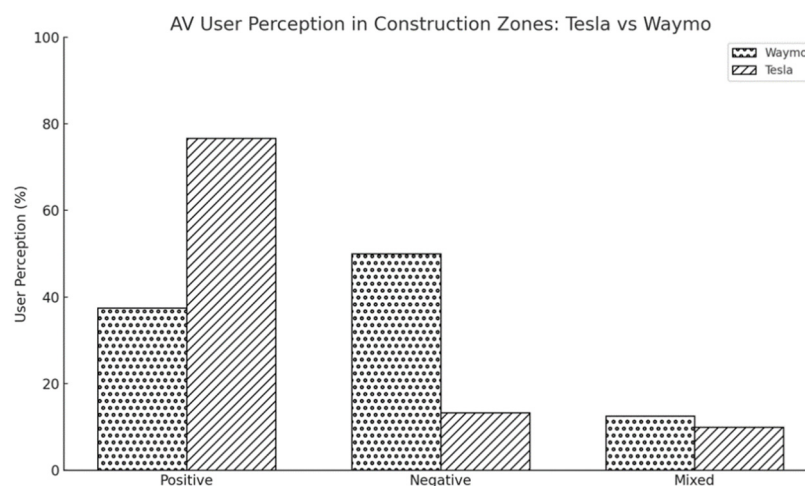


Figure 9. Comparison of user perceptions of Tesla and Waymo navigation in work zones.

4.3. Safety Implications and Regulatory Considerations of AV Failures in Work Zones

The safety implications of observed AV failures, such as misinterpretation of stop signs, failure to detect construction cones or workers, and difficulty navigating detours; warrant explicit discussion due to their potential impact on public safety and regulatory oversight. AV misbehavior in construction zones can introduce serious real-world risks. For instance, incidents where AVs failed to stop at red lights or exceeded reduced speed limits in work zones [19] highlight the danger posed to both road workers and other road users. Similarly, Tesla FSD’s confusion when interpreting a worker’s stop sign, or hesitation near pedestrian activity [36], could result in unintended vehicle movement that jeopardizes human safety. These failures are not just technical issues; they are potential regulatory triggers. In many jurisdictions, AVs are expected to meet stringent functional safety standards and comply with local traffic laws. Disengagements requiring human override suggest that current systems fall short of SAE Level 4 autonomy expectations and may influence how policymakers classify and certify AV readiness for public roads. Moreover, regulatory agencies such as NHTSA may view persistent errors in work zones as indicative of insufficient real-world training data or poor generalization in AV algorithms. These shortcomings could necessitate new guidelines around AV testing and require manufacturers to demonstrate system robustness in temporary and irregular environments. The implications also extend to liability; if an AV fails to detect a worker or misreads a stop sign leading to a collision, questions of fault, whether attributable to the manufacturer, operator, or infrastructure, become legally and ethically complex. Therefore, addressing these challenges through improved AV perception models, clearer infrastructure standards, and cross-sector collaboration is vital.

5. Research Findings

This study set out to examine the performance of AVs in navigating work zones by analyzing real-world user-documented experiences shared on YouTube. Between 2018 and 2024, AV navigation in work zones evolved from a relatively limited area of focus into a prominent subject of documentation and analysis. Through a comprehensive assessment of user narratives, on-screen captions, subtitles, and transcribed observations, the research highlighted the progress and limitations of AV systems developed by Tesla and Waymo when confronted with the construction environments. A distinctive value of this study lies in its use of user-generated content, which not only documented AV performance but also captured the emotional and cognitive responses of AV users. These insights offered a more complete evaluation of AV deployment by revealing the “human factor” behind

technological performance. These first-hand narratives offered insights into moments of confidence, such as when an FSD system navigated a complex cone layout with limited human intervention, and moments of anxiety or loss of trust, such as when the system followed a construction worker too closely or came to an unexplained stop in a confusing detour, which required human assistance either remotely or on-site. The observations revealed how users perceive, interpret, and respond to AV behavior in real-world contexts. Real-world demonstration tests are often accompanied by post-experiment surveys, while traffic simulation studies are frequently paired with driving simulator experiences to assess the human comfort level and perception on automated systems. This combined approach enables researchers to capture insights across three critical dimensions: system performance, infrastructure interaction, and rider comfort. Similarly, analyzing user-generated YouTube content in our study served a comparable purpose, which offered a window into how AVs perform in real-world settings, how they interact with road infrastructure, and how comfortable or confident passengers and bystanders feel during these encounters.

To support reproducibility and enable other researchers to conduct similar studies, we propose a standardized framework for annotating AV behavior using public video content. Our approach involved collecting user-generated YouTube videos through a custom Python script that interacted with the YouTube Data API, applying Boolean keyword combinations such as “construction zone” and “self-driving” across a defined date range, as detailed in Table 1. Videos were retained if they were created by users; and included narration, captions, or subtitles, ensuring sufficient contextual information for analysis. AV behavior within these videos, was annotated as either a “Strength” or “Challenge” was classified into six thematic categories: Perception & Sensing, Decision-Making & Planning, Control & Speed Management, Human Interaction & Override, Infrastructure Compliance & Navigation, and Environmental & Weather Conditions. These categories were established using the OpenAI API to assist with thematic clustering and pattern recognition, followed by manual validation to ensure accuracy and consistency. The comparison between automated and manual classifications yielded an accuracy of 88.67% across 222 annotated events, demonstrating strong alignment between model-assisted and human-coded results.

From this analysis, a total of 117 strengths and 105 challenges were identified across 38 user-documented AV operation cases in work zones spanning different locations. These findings indicate that AVs demonstrate a balanced but still-developing performance in complex and irregular road environments. While these strengths highlight meaningful progress toward greater autonomy, they are not sufficient on their own to achieve SAE Level 4 or 5 autonomy. Tesla’s Full Self-Driving system offers a range of advanced driver assistance features, while still requiring continuous driver supervision and responsibility in the work zone. Waymo’s autonomous driving system is designed to operate vehicles independently within specific environments. While it does not rely on a safety driver during deployment, real-world observations indicate that it occasionally required remote or on-site assistance when navigating complex or irregular road conditions. While the naming of such systems may reflect intended future capabilities, their actual functionality corresponds to differing levels of automation when viewed through the lens of the SAE automation framework, and neither system meets the criteria for full autonomy as defined at Level 5, which entails operation under all roadway and environmental conditions without human oversight.

To further contextualize these findings, user perception of AV behavior in work zones was analyzed. Approximately 68% of documented reactions were positive, 21% were negative, and 11% were mixed. The analysis of specific AV capabilities showed that AV perception and sensing were the most frequently referenced aspects, with 30 strengths and 30 challenges. This balance reflects ongoing inconsistencies in how AVs interpret cones, lane

markings, signage, and nearby actors. In contrast, decision-making and speed modulation functions showed greater reliability, with more strengths than challenges. Notably, human interaction and override functions emerged as key weaknesses, having the lowest strength counts and the highest challenge frequency, underscoring the complexity of engaging effectively with human agents in dynamic work zone environments.

Expanding this comparison, insights from users also shed light on the differences between Tesla and Waymo systems. Tesla's supervised FSD system exhibited strengths in real-time learning, maneuver execution, and adaptability to irregular geometry, though it was more prone to perception-related challenges and required more frequent driver intervention, likely due to its reliance on a camera-only vision system. Waymo, which employs LiDAR, radar, and cameras, demonstrated fewer perception errors and improved handling of human actors but was occasionally less flexible in responding to unmapped detours or temporary infrastructure changes. User perception was more positive for Tesla (77% of reactions) than Waymo (38%), possibly influenced by Tesla's semi-supervised mode allowing safety drivers to intervene, while Waymo's fully driverless model exposed the system to more critical observation during unexpected events. These system-specific insights were reinforced by a number of observed technical challenges. Common limitations included misinterpretation of temporary or inconsistently placed cones and barriers, difficulty detecting or reacting to faded or occluded lane markings and signage, and failure to interpret pedestrian gestures or construction worker hand signals in certain instances. Furthermore, some systems exhibited hesitation or inability to reroute around temporary detours or physical work zone obstructions. In parallel, infrastructure-related challenges emerged as a significant contributor to AV disengagements. These issues were frequently associated with zones featuring nonstandard geometry, inadequate signage, or ambiguous lane boundaries. AV confusion was prevalent in environments lacking high-contrast or reflective materials, or where cones were irregularly placed, especially under low-light or visually obstructed conditions. These findings emphasize the interdependence between AV design and infrastructure readiness.

To contextualize these findings, the 2018 NHTSA report [56] on Automated Driving Systems (ADS) presented performance metrics from public road testing that closely align with the challenges identified in this study. The report documented safety-critical events, including crashes and disengagements, most commonly occurring in complex environments such as intersections and non-standard road configurations. These conditions are consistent with the work zones analyzed in this study, where AVs exhibited uncertain or inconsistent behavior. Furthermore, the NHTSA's emphasis on the need for a "fallback-ready user" reflects this study's observations of frequent human intervention and user discomfort when AV systems failed to confidently navigate work zones. Furthermore, the findings of this study are well-aligned with the priorities and concerns highlighted in the 2021 final report of Autonomous/Connected Vehicles Assessment study in Work Zones [57]. First, the frequent user-reported confusion and system disengagements in work zones reflect the report's identification of work zones as high-risk, complex environments requiring targeted AV adaptation and infrastructure support. Second, users' calls for clearer communication, signage, and system transparency reinforce the report's emphasis on improving human-machine interface design and driver awareness. Third, the paper's insight into mixed public perception echoes the report's recognition of the importance of public education, outreach, and trust-building. Lastly, both our study and the report [57] stress the need for continuous learning from real-world deployments, stakeholder feedback, and collaborative efforts to advance AV safety and integration. Additionally, the 2024 report by the U.S. Department of Transportation [58] reinforces our findings by identifying construction zones as a key non-ADS factor contributing to disengagements, specifically

citing scenarios such as blocked lanes. This consistency highlights a shared understanding that current AV systems face significant challenges operating reliably in construction zones, underscoring the need for enhanced system adaptability, improved infrastructure support, and targeted safety measures in these complex environments.

6. Future Directions

In response to the combined technical and infrastructure challenges, several targeted directions for improvement are proposed. Infrastructure enhancements should prioritize high-contrast, retroreflective, and machine-readable elements that function under varying weather and lighting conditions. Integrating Bluetooth beacons and dynamic signage could also help guide AVs during temporary layout changes. On the technology front, efforts should focus on strengthening perception models to recognize temporary and occluded signage, improving the recognition of human gestures, and enhancing system adaptability in diverse environmental conditions, including inclement weather. Complementing these technical and infrastructural strategies, policy and user-centered considerations must also be addressed. Regulators should develop and enforce AV-compatible work zone design standards and require manufacturers to report detailed system performance data under diverse scenarios. In terms of rider experience, incorporating real-time, context-aware in-vehicle messages, such as “Yielding to road worker” or “Construction detected; adjusting path”; can help reassure passengers and build trust. Structured educational tools, including mobile apps or onboarding sessions, can further support user understanding of AV behavior, particularly in work zone scenarios. To support safe and consistent AV operation in dynamic environments, improvements in human interaction protocols are also essential. Training for construction personnel on standardized AV interaction cues and emergency disengagement procedures could foster safer and more predictable engagements between AVs and human workers. Furthermore, enhancing data diversity is crucial. Future research should include a broader range of AV manufacturers, geographic regions, and digital platforms beyond YouTube to improve the generalizability of findings. Building on the insights identified in this study, future work will also focus on grounding observed AV decision-making in construction zones within formal algorithmic frameworks. Specifically, AV behaviors will be analyzed through the lens of established planning and optimization strategies, such as A^* , D^* , RRT, and constrained decision-making models. This approach will enable stronger connections between real-world AV performance and theoretical models of decision-making under uncertainty.

7. Limitations

While the findings from this study offer valuable insights, several limitations must be acknowledged. The reliance on user-generated YouTube videos, while offering real-world depth, may introduce selection bias or subjective interpretations. The dataset predominantly featured videos recorded in clear daylight conditions, with limited representation of challenging conditions such as fog, snow, rain, or high wind, as well as nighttime driving. The absence of internal AV data, such as sensor inputs, decision logs, and software versions, also limited the ability to determine the precise causes of system behavior.

Additionally, user demographic and psychological variables (e.g., age, gender, driving experience, stress level, familiarity with AVs) were not captured, though they likely influence the way AV behavior is perceived and reported. In addition, the sample of YouTube content creators and viewers may not be representative of the broader public. Individuals who post or engage with AV-related content on YouTube may differ in terms of technological interest, risk tolerance, communication style, or geographic and socio-economic

background. These factors could influence which AV experiences are shared and how they are interpreted, potentially introducing bias in the observed patterns.

Another limitation lies in the geographic concentration of the dataset, which is predominantly composed of videos from the USA. Road infrastructure, signage conventions, construction practices, and driver behavior can vary across countries; all of which influence how AVs perceive and respond to work zone environments and shape how users interpret and evaluate AV performance in such contexts. For instance, a maneuver perceived as cautious or appropriate in one country might be viewed as confusing or unsafe in another due to differing norms and expectations. Expanding future datasets to include more diverse international content would enhance both the global relevance of AV behavior analysis and our understanding of cross-cultural differences in user perception.

In addition to expanding data diversity, future research could benefit from a hybrid approach that combines annotated AV-manufacturer footage with user-generated videos. In this study, we intentionally focused on user-generated videos to prioritize authentic, real-world experiences that reflect user interactions and spontaneous AV behavior. While manufacturer-produced content often offers structured insights into system behavior during edge cases, such as rare failure modes or complex construction scenarios; user-generated footage provides authentic context around real-world usage and human responses. Integrating both sources would allow for a more comprehensive analysis that leverages the technical precision of controlled tests and the experiential depth of spontaneous, real-world encounters. This blended methodology contributes to a more holistic evaluation of system robustness, safety, and user trust.

Given these limitations, future research should pursue several important directions. These include comparing the infrastructure and technological adaptations needed for safe AV navigation in rural versus urban areas, as well as exploring work zone personnel's perceptions of AV operations. Further investigation is also needed to assess the effectiveness of external AV indicators, such as lights or signage; in improving worker trust and safety. Additionally, studying AV performance under adverse weather conditions in conjunction with complex work zone layouts will be critical. Finally, expanding the dataset to include insights from AV users in cold-climate regions may inform strategies for safe navigation during road closures and hazardous weather events.

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